Introduction to Speech and Language Processing

Practical Assignment #3 - Dialog Systems

UA Campus Assistant

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1 Introduction

The UA Campus Assistant is a task-oriented spoken dialogue system (SDS) produced to fulfill the requirements of IPFL course.

This dialog system was designed to cover tasks grounded to University of Aveiro context. Tasks include schedule queries, nearest canteen lookup, canteen menu, exam information, and exam date saving.

The system is implemented using Rasa Open Source 3.6, Whisper ASR, and gTTS TTS. It applies SDS concepts learned in the course: frame-based dialogue management, 4-tiered confidence policy, explicit/implicit confirmation, and grounding.

2 System Architecture

The system follows the standard SDS pipeline architecture (Figure 1). Table 1 shows the role of each component and what technology is used for each implementation.

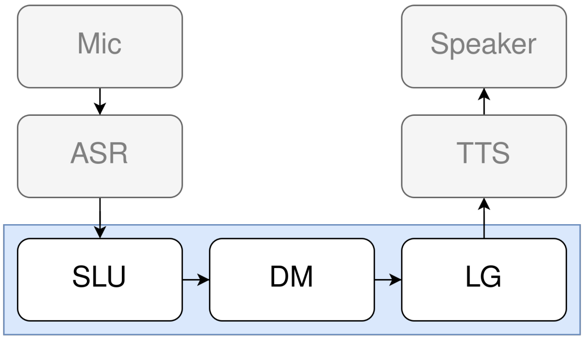


Figure 1: Pipeline architecture.

Comp.	Role	Implementation
ASR	Speech-to-text	Whisper (base.pt)
SLU	Intent + entities	Rasa DIET
DM	State & policy	Rasa Core
LG	Response generation	`utter_*` & templates
TTS	Text-to-speech	gTTS

Table 1: SDS pipeline components.

3 Frame-Based Dialogue Management

In frame-based dialogue management, each task the system can perform is represented as a *frame*, a structured record with a set of typed slots that must be filled before the task can execute. The system's goal is to populate the frame for the active task by eliciting missing information from the user.

The system defines five frames, each implemented as a Rasa *form*. A form activates when the triggering intent is detected and remains active until all required slots are filled, prompting the user for each missing value in sequence. Optional slots are populated only if the user provides the corresponding entity spontaneously.

Frame	Required Slots	Optional Slots	Rasa Form
Query schedule	`day`	`time_of_day`, `time`	`schedule_form`
Nearest canteen	`location`	—	`canteen_form`
Query menu	`canteen`, `meal_type`	`query_date`	`menu_form`
Query exam	`subject`	—	`exam_query_form`
Save exam	`subject`, `exam_date`	`exam_time`, `exam_room`	`save_exam_form`

Table 2: Frames and their slots. Required slots are elicited if absent; optional slots are filled opportunistically.

The system implements **mixed initiative**, which means that a fixed question order is not enforced. If the user provides multiple slot values in a single utterance (e.g., *"Que aulas tenho segunda de manhã?"*), the NLU extracts both `day=segunda` and `time_of_day=manhã` simultaneously, and the form skips the corresponding prompts entirely. This allows experienced users to complete a task in a single turn, while the system still guides novice users step by step.

4 Dialogue Acts

The system defines 14 intents mapped to dialogue acts (following Verbmobil taxonomy):

Intent	Dialogue Act	Example utterance
`greet`	GREET	*"Olá"*
`goodbye`	BYE	*"Adeus"*
`thank`	THANK	*"Obrigado"*
`ask_schedule`	REQUEST	*"Que aulas tenho segunda?"*
`ask_nearest_canteen`	REQUEST	*"Cantina mais próxima do DETI?"*
`ask_menu`	REQUEST	*"Ementa de almoço em Santiago?"*
`ask_exam_info`	REQUEST	*"Quando é o exame de IPFLN?"*
`ask_known_locations`	REQUEST-COMMENT	*"Que locais conheces?"*
`save_exam_date`	INIT	*"Guardar exame de física"*
`inform`	FEEDBACK	*"Estou no DETI"*
`affirm`	ACCEPT	*"Sim"*
`deny`	REJECT	*"Não"*
`ask_identity`	REQUEST-COMMENT	*"Quem és?"*
`out_of_scope`	—	*"Como está o tempo?"*

Table 3: Intent-to-dialogue-act mapping.

5 Dialogue Management Components

The system evaluates NLU confidence c before activating forms and applies a four-tier policy:

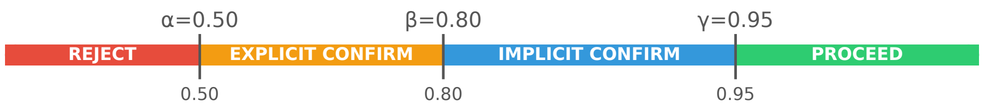
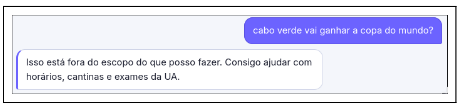


Figure 2 shows live examples of each behaviour. Note: thresholds were temporarily adjusted ($\beta' = 0.97$, $\gamma' = 0.999$) to expose all tiers during demonstration, since the well-trained DIET classifier produces high-confidence predictions on common phrasings.



(a) REJECT: out-of-scope input

(b) EXPLICIT CONFIRM: low confidence

(c) IMPLICIT CONFIRM: slot echoed in next question

(d) PROCEED: direct response, no confirmation

Figure 2: Four tiers of the confidence policy observed in the running system.

Grounding. The `inform` intent triggers `utter_hmhm` (*"Ok."*, *"Certo."*, *"Entendido."*) as shown in Figure 3. Implicit confirmation (tier 3) is itself a form of display grounding: the system echoes the understood slot value inside the next question without requiring an explicit yes/no.

Figure 3: Grounding tokens used to acknowledge user input.

The `ask_identity` intent triggers `utter_identity`, which explicitly identifies the system as a chatbot: *"Não sou humano, sou um chatbot académico."* providing **transparency** to the user.

6 Language Generation

The system uses a two-tier LG strategy:

`utter_*` templates: used for all control-flow responses (e.g., slot prompts, confirmations, rejections, etc.). Some templates include slot placeholders filled by Rasa at runtime:

```
utter_confirm_past_exam:
  - text: "Atenção: {pending_exam_date} já passou. Queres mesmo guardar?"
```

Custom action NLG: content-bearing responses are assembled in Python inside custom actions. The action formats the retrieved data into a structured fallback string. If the `ENABLE_GROQ_RESPONSE` flag is set, this fallback is passed to an *LLM* via the Groq API, which produces a more natural phrasing in Portuguese while being strictly constrained to the provided data.

7 Evaluation

7.1 NLU: Intent Classification

The NLU component was evaluated on a held-out test set of 63 examples (never seen during training), covering all 14 intents. Overall accuracy was **93.7%** (59/63 correct). Figures 4 and 5 show confusion matrix and per-intent F1 scores, respectively.

Error analysis. Four errors were observed: `save_exam_date` (2 errors) is semantically close to `ask_exam_info` (both involve an exam subject). `goodbye` and `deny` each have one informal utterance that the model misclassifies due to limited coverage of colloquial expressions.

7.2 Dialogue: End-to-End Demonstrations

Figure 6 shows two complete multi-turn conversations. Conversation (a) exercises the `save_exam_form`: the system detects an existing exam entry and asks for explicit confirmation before overwriting it, then immediately answers a follow-up query using the saved data. Conversation (b) exercises grounding and two consecutive `canteen_form` activations: after the first lookup the user provides an `inform` utterance changing their location, the system acknowledges with a grounding token (*"Certo."*) and correctly resolves the second nearest-canteen query, followed by a menu lookup via `menu_form`.

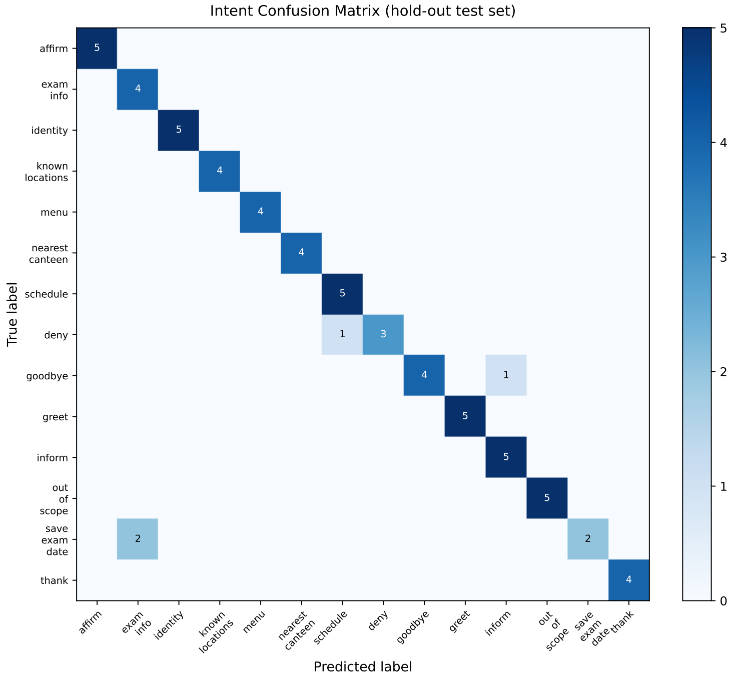


Figure 4: Intent confusion matrix (hold-out test set, $n = 63$).

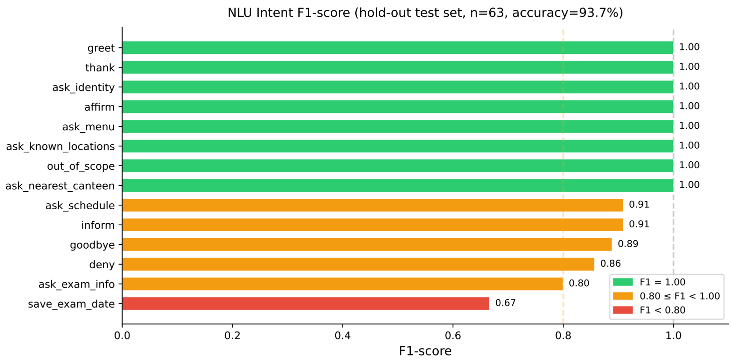


Figure 5: Per-intent F1 scores on the hold-out test set.

(a) Exam save with update confirmation + query

(b) Nearest canteen (×2, location update) + menu

Figure 6: End-to-end dialogue demonstrations.

8 Conclusion

The UA Campus Assistant implements a complete SDS pipeline with frame-based dialogue management, 4-tiered confidence policy, explicit and implicit confirmation, acknowledgement grounding, and AI transparency disclosure. NLU intent accuracy reached 93.7% on unseen data, with the remaining errors concentrated in semantically overlapping intents. Even though the accuracy is high, the real value is shown through the demo that showcases the assistant's capabilities in real-world scenarios.

The integration with **Groq** makes the responses more fluent and natural, enhancing the user experience.

This project helped the student understand in practice how conversational agents can be built using a combination of rule-based and machine learning approaches, while discovering the challenges involved in building conversational agents.

Future Work

Two directions stand out. First, **bilingual PT+EN support**: adding English training examples for each intent and a `language` slot detected on the first turn would allow the system to serve both Portuguese students and international students naturally. Second, **expanding the test set**: the current NLU test set contains 63 examples; a larger and more diverse set (covering informal phrasings, code-switching, and ASR-transcribed utterances) would yield more reliable accuracy estimates and expose the remaining intent confusion.